

Morphological Filter

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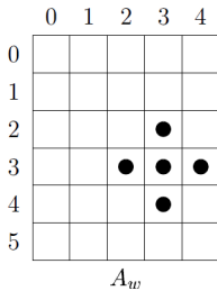
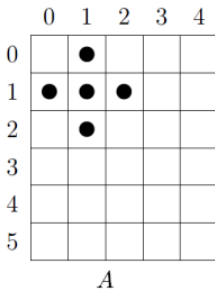
- Originally operated on Binary (black and white) image
- Morphological filters alter local structures in an image

Translation

A is set of pixels in binary image

- $w = (x, y)$ is a particular coordinate point
- A is set A “translated” in direction (x, y) . i.e

$$A_x = \{(a, b) + (x, y) : (a, b) \in A\}$$

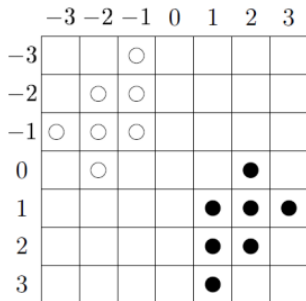


Reflection

A is set of pixels

- Reflection of A is given by

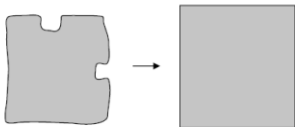
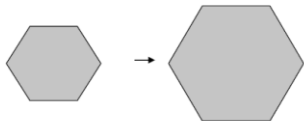
$$\hat{A} = \{(-x, -y) : (x, y) \in A\}$$



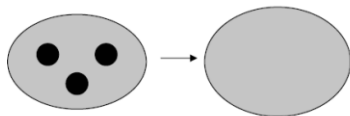
- 2 basic mathematical morphology operations, (built from translations and reflections)
 - Dilation
 - Erosion
- Also several composite relations
 - Closing and Opening
 - Conditional Dilation

- Dilation expands connected sets of 1s of a binary image. It can be used for

1. Growing features

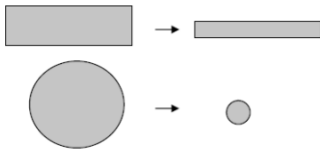


2. Filling holes and gaps

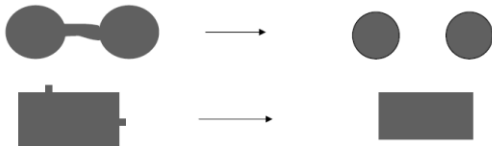


- Erosion shrinks connected sets of 1s in binary image. Can be used for

1. shrinking features



2. Removing bridges, branches and small protrusions



The Structuring Element

- A structuring element is a shape mask used in the basic morphological operations
- They can be any shape and size that is digitally representable, and each has an origin.



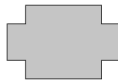
box



hexagon

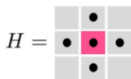


disk



something

$$H(i, j) \in \{0, 1\}$$



$H =$

 origin (hot spot)

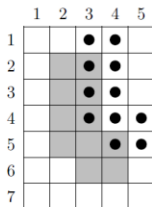
- Morphological operation can be expressed by describing images as 2D point sets
- For example, for binary images

$$Q_I = \{p : I(p) = 1\}$$

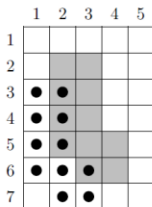
Dilation

- Suppose A and B are sets of pixels, dilation of A by

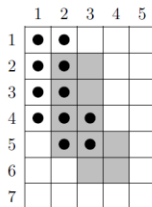
$$A \oplus B = \bigcup_{x \in B} A_x$$



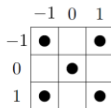
$A_{(-1,1)}$



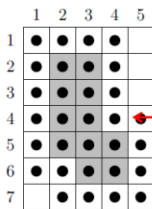
$A_{(1,-1)}$



$A_{(-1,-1)}$



B



$A \oplus B$

Union of all translations

Dilation Example

Cross-Correlation Used
To Locate A Known
Target in an Image

Text Running
In Another
Direction

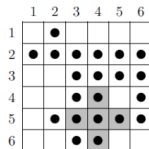
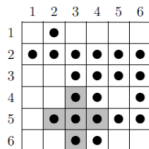
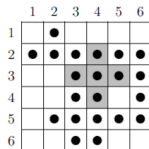
**Cross-Correlation Used
To Locate A Known
Target in an Image**

**Text Running
In Another
Direction**

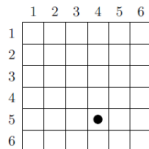
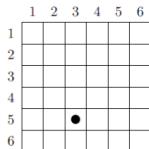
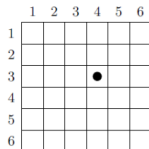
Erosion

- Given sets A and B, the erosion of A by B: find all occurrences of B in A

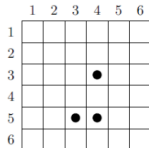
All occurrences
of B in A



For each
occurrences
Mark center of B



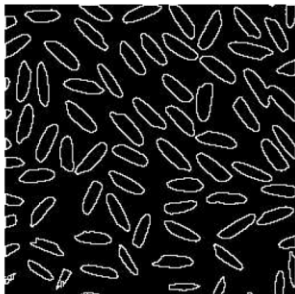
Erosion: union
of center of all
occurrences of
B in A



$$A \ominus B$$

Boundary Detection

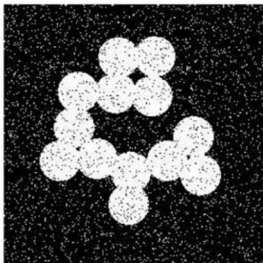
$$A - (A \ominus B)$$



- Opening and closing: operations built on dilation and erosion
- opening = erosion followed by dilation
- closing = dilation followed by erosion

Noise Removal: Morphological Filtering

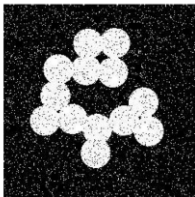
- Suppose A is image corrupted by impulse noise (some black, some white pixels, shown in (a) below)



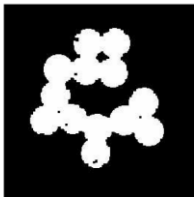
(a)

- $A \ominus B$ removes single black pixels, but enlarges holes
- We can fill holes by dilating twice $((A \ominus B) \oplus B) \oplus B$

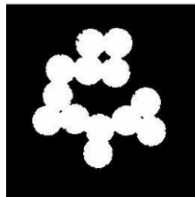
Noise Removal: Morphological Filtering



(a)



(b)



(c)

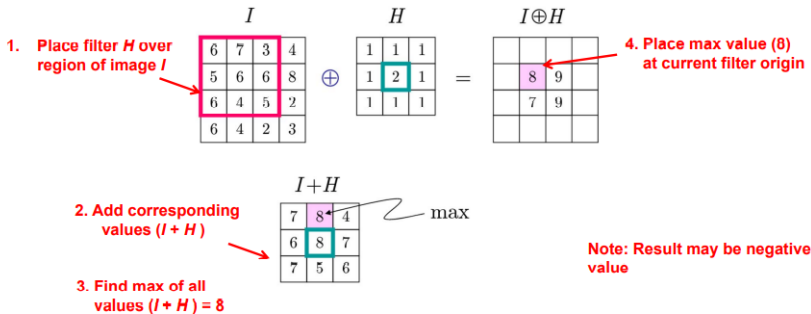
- First dilation returns the holes to their original size
- Second dilation removes the holes but enlarges objects in image
- To reduce them to their correct size, perform a final erosion:
$$(((A \ominus B) \oplus B) \oplus B) \ominus B$$
- Inner 2 operations = opening, Outer 2 operations = closing
- This noise removal method = opening followed by closing
$$(A \circ B) \bullet B$$

- Morphology operations can also be applied to grayscale images
- Just replace (OR, AND) with
- For color images, perform grayscale morphology operations on each color channel (RGB)
- For grayscale images, structuring element contains real value

Grayscale Dilation

Grayscale dilation: Max (value in filter H + image region)

$$(I \oplus H)(u, v) = \max_{(i,j) \in H} \{I(u + i, v + j) + H(i, j)\}$$



Grayscale Erosion

Grayscale erosion: Min (value in filter H + image region)

$$(I \ominus H)(u, v) = \min_{(i,j) \in H} \{I(u+i, v+j) - H(i,j)\}$$

